

MSc AIML Sem I SCM

Assignment 1 (Errors, Bisection, False Position, Secant, Newton Raphson Methods)

1. Explain the different types of errors that occur during computation
2. Find an interval of unit length which contains this root of $f(x) = x^3 - 9x + 1 = 0$.
3. Find the root of the following using Bisection Method upto 1 decimal places
 $f(x) = x^3 - 15.2x + 13.2$, the root lies between 0 and 1
4. Find the root of the following using False Position Method upto 2 decimal places
 $(x) = x^4 - x - 10$, the root lies between 1.8 and 2
5. Find the root of the following using Secant Method upto four decimal places:
 $x^3 - 2x - 5$ starting with 2, 3
6. Find the root of the following using Newton Raphson Method:
 $x^3 - 2x - 5$ to four decimal places starting with $x_0 = 2$
7. Explain the different stopping criteria for root finding methods

Answers

2. [2,3]
3. 0.91406
4. 1.85558
5. 2.09455
6. 2.09455